Highlights

Statistical Validation-Driven Performance Analysis of Forensic Imaging Tools in USB Forensics: Evaluation of Imaging and Hashing Across Diverse Data Sets

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- A comprehensive investigation compares FTK Imager and OSForensics with regard to imaging speed, hashing accuracy, and resource consumption for large-scale digital forensic enquiries.
- Statistical validation employing Shapiro-Wilk, Mann-Whitney U, and Spearman correlation tests reveals significant performance disparities and identifies a crucial hashing anomaly.
- Practical forensic dataset analysis offers optimization strategies to enhance imaging efficiency and reliability in resource-constrained, large-scale environments.

Statistical Validation-Driven Performance Analysis of Forensic Imaging Tools in USB Forensics: Evaluation of Imaging and Hashing Across Diverse Data Sets

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Abstract

Due to the complexity and scale of cybercrimes, reliable forensic procedures are in great demand. Digital forensic technologies capture and preserve court-admissible evidence. Due to their portability and large storage capacity, USB devices are widely used, putting individuals and organizations at risk of data theft. As forensic imaging tools play a vital role in digital investigations, a methodological approach is needed to guarantee performance reliability, resource efficiency,and data authenticity. This paper evaluates FTK Imager and OSForensics, two popular forensic imaging tools, for speed, hashing accuracy, resource utilization, and statistical significance. The study examines how resource efficiency impacts forensic imaging in high-volume, time-sensitive environments by analyzing tool performance dynamics. Performance flow is analyzed to determine efficiency metrics such as CPU and memory usage. FTK Imager outperformed OSForensics in CPU and memory efficiency, improving it for resource-constrained environments. The Shapiro-Wilk test showed non-normal data distribution, requiring non-parametric tests. The Mann-Whitney U test showed statistically significant differences in imaging speeds. Spearman correlation analysis showed a strong positive relationship between CPU utilization and imaging speed, showing how resource usage affects performance. A notable anomaly was detected when OSForensics generated identical MD5 and SHA-1 hashes for two USB drives, unlike FTK Imager, which generated unique hashes. This raises questions about imaging hashing reliability and error propagation. To demonstrate practical implications, a forensic dataset analysis is included, which demonstrates the way to optimize performance and reliability. This research contributes to advancing forensic imaging by emphasizing the importance of statistical validation and resource efficiency in large-scale investigations.

Keywords: USB Forensics, Digital Forensic Imaging, Hashing, FTK Imager, OSForensics

1. Introduction

In the last ten years, the generation of digital data, especially personal and sensitive information, has increased significantly across many platforms and devices. This has increased global awareness regarding data security, privacy, and ethical data management to prevent identity theft and fraud. Digital forensics has progressed from fundamental recovery to sophisticated investigations involving networks, mobile devices, and cloud settings. Maintaining the integrity of digital evidence and a clear chain of custody is crucial. (1). Digital Forensic tools obtain, process, and examine evidence, which must be accurate and reproducible for judicial proceedings. However, the domain is deficient in strong validation standards, which raises concerns over the accuracy and reliability of forensic tools—particularly as the demand for ISO 17025 accreditation escalates. (11).

File System Forensics provides a comprehensive guide on the operations of several file systems, including FAT, ExFAT, and NTFS, and the associated digital forensic equipment (18) . Imaging and hashing are essential elements of digital forensics. Imaging produces a sequential replica of storage media to safeguard original evidence (2), whereas hashing generates a distinctive digital fingerprint to ascertain integrity(3). These measures are essential to protect the chain of custody and guarantee admissibility.

USB forensics has identified detachable media as a common tool for data theft and criminal activities because to its portability and large storage capacity (4). USB drives are used by cybercriminals to steal data, transmit malware, and access confidential data. Due to their portability and usability, these devices are often used to hide illegal data or bypass security. Having excellent forensic tools to scan and hash USB devices is crucial for extracting evidence quickly and accurately in digital investigations (5).

As forensic workloads increase, delays in imaging or hashing caused by resource limitations may hinder time-sensitive investigations. Maintaining high-speed performance and reliability is an ongoing challenge. A thorough forensic tool performance evaluation is needed to identify bottlenecks and improve efficiency. Digital forensics uses software tools to capture and process digital evidence, but the validation of these tools is insufficiently addressed, with minimal empirical study into their reliability and efficacy. Although practitioners are responsible for deploying these technologies accurately and effectively, inherent tool flaws and understanding their limitations can lead to user errors and legal uncertainty.

1.1. Motivation

Due to the increase of digital data and cybercrime complexity, reliable digital forensic tools are required. Image and hashing determine legal admissibility and evidentiary integrity. FTK Imager and OSForensics are widely used despite limited empirical research on their performance. Examining forensic tools' efficacy on such media is vital given the increasing dependence on USB devices in cyber events. This study compares FTK Imager to OSForensics in imaging, hashing, and resource use. Strong statistical techniques identify performance bottlenecks and provide recommendations for tool selection in actual investigations.

1.2. Contribution

This subsection highlights the key contributions of this research, emphasizing its relevance in the field of digital forensics.The study seeks to address significant shortcomings by delivering a thorough performance evaluation of two prevalent forensic tools — FTK Imager and OSForensics. The contributions are summarized as follows:

- **Empirical Performance Analysis:** compares imaging and hashing speeds to determine which tool excels under different conditions.
- **Resource Utilization Metrics:** Assess CPU and memory usage as a measure of hardware efficiency during forensic operations.

- **Statistical Validation**: Tests for significant performance differences, expanding the depth of analysis.
- **Identifies anomalies:** Identifies issues such as OSForensics generating identical hash values for two files, raising reliability concerns.
- **Practitioner Guidelines:** Helps digital forensic investigators choose appropriate tools for specific tasks.

Table 1: Summary of Research Questions, Objectives, and Corresponding Sections

Research Question	**Objective**	**Section Addressing the Question**
What are the differences in imaging and hashing speeds between FTK Imager and OSForensics across various USB drive sizes and file systems?	To compare the performance of FTK Imager and OSForensics in terms of imaging and hashing speeds.	Results and Discussion
How do FTK Imager and OSForensics compare in terms of CPU and memory utilization during forensic imaging and hashing?	To analyze resource utilization metrics (CPU and memory) during forensic imaging and hashing.	Results and Discussion
Is there a statistically significant difference in performance metrics between the two tools?	To determine whether performance differences are statistically significant.	Statistical Analysis in Results and Discussion
Which tool demonstrates superior resource efficiency under varying operational conditions?	To evaluate the overall efficiency of each tool under various scenarios.	Comparative Analysis in Results and Discussion

Table 1 outlines the research questions, objectives, and the respective sections where each is addressed, providing a structured roadmap of the study's contributions.

The remainder of this paper is organized as follows:

- Section 2: Related Work - Reviews existing research related to forensic imaging and hashing tools.
- Section 3: Experiment Design - Describes the experimental setup, tools, and performance metrics used in the study.
- Section 4: Results and Discussion - Presents the experimental findings and discusses their implications.
- Section 5: Conclusion and Future Scope-Summarizes key findings, study limitations, and future research directions.

2. Related Work

There is an increasing demand for digital forensic investigators, caused by the increasing size of data targets, which existing forensic tools find difficult to process efficiently. To address these issues, the authors of (6) recommend optimizing software design, using multithreading on contemporary multicore CPUs, and implementing distributed processing, particularly taking advantage of commodity GPUs for computational tasks such as binary string searches. By showing how GPU offloading can greatly improve performance, the authors argue for the creation of GPU-aware digital forensics software to help with analysis in a world where data is getting more complicated.

An innovative methodology for robust performance testing of digital forensic tools is presented in paper (7). The article emphasizes the need for multiple observations and the use of orthogonal arrays (OAs) to improve testing efficiency and reduce test cases. This adaptive experimental strategy enhances performance testing quality and enables automation, potentially replacing human testers in complex forensic settings. Although software behavior is unpredictable and universal testing metrics are challenging, the authors suggest peer review to establish new standards in the digital forensic community.

A study by (4) analyzes registry and log data before, during, and after USB insertion to examine Windows 10 device usage timing and context. Windows 10 USB forensic analysis saves digital evidence from MSC-, PTP-, and MTP-enabled device insertion and removal using the registry and Windows Event Viewer. Research suggests using the registry's DeviceClasses and USB standards to extract USB devices' first insertion timestamps. For thorough forensic investigations, Windows Event Viewer operational events in the Microsoft Windows/WPD-MTP ClassDriver path provide USB insertion and removal timestamps.

According to (8), there is a significant research gap in assessing forensic imaging techniques for second-hand storage devices in New Zealand. This highlights the need for comprehensive analysis of these technologies in forensic investigations. The authors used quantitative research to assess DC3DD, DCFLDD, and Guymager's performance against NIST's functional and optional requirements. They found that DC3DD and Guymager mostly met the criteria, but Guymager had a better GUI and better log data, while DC3DD relied on command-line inputs. A high percentage of the hard drives examined contained sensitive personal and organizational data, indicating a worrying lack of data security awareness among local users.

In (9), residual data, including personally identifiable information (PII) on open-market reconditioned hard drives is rigorously evaluated, revealing significant data sanitization standards gaps among enterprises. Despite GDPR and NDPR, data disposal's technical responsibilities are poorly understood, posing legal and ethical risks, particularly regarding the unauthorized retention of sensitive information. The analysis suggests stronger enforcement, industry-wide standardization, and stakeholder education on data sanitization to reduce privacy violations and maintain information governance during refurbishment.

The paper (18) recommends a systematic approach to extract file content and metadata, enabling manual validation of forensic tool results and preserving forensic investigations.

The study (10) found that while disk-to-file imaging with cryptographic hash verification is standard, the expected variation in advanced features among tools is unsupported, as most tools have comparable feature sets with minimal innovation. The discovery that many of these tools have not been updated in the past year indicates a significant maintenance and optimization gap, suggesting that current tools may not be enough to address digital investigators' changing needs. This stagnation in tool development highlights the need to prioritize enhancing essential forensic tools to improve digital investigations.

A critical analysis of tool-testing methodologies in digital forensics in 2018 is presented in (11). It highlights the challenges of scientific integrity and reliability in forensic software. In addition, the article presents survey data showing practitioner consensus on these issues.

(12) analyzed 800 scholarly articles from 2014-2019 to evaluate digital forensic tools, focusing on availability, maintenance, and documentation. The study found 62 digital forensics tools, but only 33 were publicly available, with many lacking ongoing support once developed. They propose a central repository for rigorously tested digital forensic tools to fix current issues and improve the landscape. They advise developers to write comprehensive code documentation and create modular plugins instead of standalone apps to improve interoperability and usability.

In (13), a new method for recovering deleted files in NTFS file systems is introduced using Python and PyTSK3 to reconstruct data using the Master File Table (MFT). The digital forensics methodology emphasizes metadata integrity to aid recovery in complex file fragmentation and overwrite situations. The article (14) explores digital forensic tool challenges. Focuses on rigorous validation processes for forensic investigations to maintain evidence integrity and dependability. The (15) article explores the challenges of using digital forensic tools. Forensic investigations require rigorous validation processes to maintain evidence integrity and dependability.

The (16) explains data concealment strategies to educate digital forensics professionals on anti-forensic measures that obfuscate data, complicating investigations. It uses multiple detection methods and the "rfinder" application to combat anti-forensic technologies like Data Mule FS, which distort file system architectures to hide data. The discussion focuses on Ext2 and Ext3, but the concepts and strategies are applicable to other file systems like FAT and NTFS, emphasizing the need for adaptable forensic methodologies in light of evolving anti-forensic techniques.

In (20), a methodology is presented for extracting data from embedded devices, such as Internet routers, by evaluating various access techniques to access device memory. It emphasizes using hardware hacking insights to create an investigative framework that improves data acquisition efficiency and success. Under ideal conditions, it was found that, both complete file-system and physical acquisitions of the device's internal flash memory are possible, and the retrieved data can be converted into industry-standard digital forensic formats for comprehensive forensic analysis.

The work (17) examines digital tools in computer forensics. This study compares proprietary and open-source tools to find those with better metrics. Use of comparative tables based on existing literature guarantees that the chosen tools meet strict relevance, completeness, and timeliness requirements for this assessment. The methodology aims to create a comprehensive framework for evaluating these forensic tools' ability to meet investigative requirements and improve digital forensic investigation integrity.

The article (19) suggests a systematic framework for digital forensic tools to enhance validity, reliability, and standardization, particularly as automation in these tools becomes more complex. The study proposes an intermediate output format to validate monolithic forensic tools and address potential errors at each processing stage using a comprehensive process model that dissects their internal workflows. This model leads to the concept of a 'abstract digital forensic tool,' which explores process interdependencies and allows critical analysis of error propagation using a Cyber-investigation Analysis Standard Expression (CASE)-annotated dataset.

This paper (21) synthesizes web, disk, and email forensics research, highlighting best practices. The five-step forensic investigation process includes evidence identification, collection, examination, assessment, and reporting. Different forensic tools are needed for each phase. Analysis of e-commerce shopping cart dynamics, disk forensics with FTK Imager, and Gmail's original message structure show the importance of intelligent tool selection in these forensic areas and suggest future research.

A study in (22) found that Windows' Virtual Secure Mode (VSM) can cause a Blue Screen of Death (BSoD) in digital forensics when used with memory acquisition tools like FTK Imager and Belkasoft RAM Capturer. A static and dynamic code analysis reveals that while these tools remain functional under different VSM conditions, active VSM significantly increases unreadable memory, causing FTK Imager to crash in the ad_driver.sys module. VSM's impact on memory acquisition must be considered by digital forensic investigators, who may use RAM dumps in live investigations.

The paper (23) examines the reliability of memory forensics practices, examining address layout and translation, and proposes methods to improve reliability. Empirical experiments utilize FTK Imager, Magnet RAM Capture, and Belkasoft RAM Capture on Windows 10 systems.

The (24) HDD Detection and Investigation Model (DIM) employs FTK Imager forensics to detect physical damage, software errors, and malicious attacks. Design science-based DIM detection, gathering, and analysis help forensic investigators identify and assess HDD vulnerabilities in real-world scenarios. The empirical evaluation shows that the DIM improves organizational resilience against HDD threats, improving digital forensics and cybersecurity.

Digital forensic investigations require accurate file metadata interpretation, especially exFAT timestamps, which vary by operating system file system driver. According to (25), Windows, macOS, and Linux handle exFAT timestamps differently, and forensic tools such as Autopsy, X-Ways Forensics, EnCase Examiner, and FTK Imager interpret them differently Thus, forensic examiners must be careful when performing timestamp analysis because assuming file system specifications can lead to serious misinterpretations that could compromise criminal investigations.

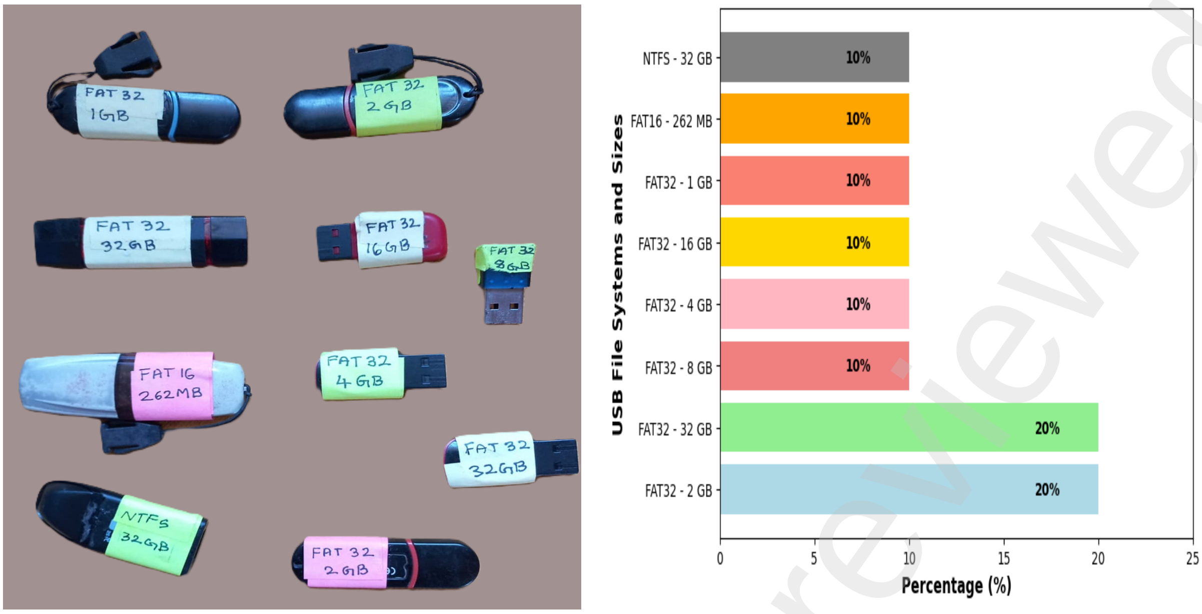
3. Experiment Design

3.1. Dataset Description

This study examined ten USB drives from different brands for variety. They vary in file system types and sizes. Each drive holds various data types, including text, video, images, audio, pdf, and ppt. The file systems considered are FAT32, FAT16, and NTFS, with sizes of 262 MB, 1 GB, 2 GB, 4 GB, 8 GB, 16 GB, and 32 GB They are numbered PD 001 to PD 010. The drives used in this study are depicted in 1(a) and their distribution in 1(b).

3.2. Testing Setup

The forensic imaging and analysis were conducted using a specialized forensic workstation, equipped with the hardware and software specifications detailed in Table2. The workstation was chosen to guarantee rapid processing, effective forensic imaging, and real-time performance assessment.



(a) (b)

Figure 1: Dataset Details: (a) USB Drives used (b) Distribution of Dataset

Table 2: Hardware, Software, and Experimental Environment Configuration

Category	**Details**
Hardware Configuration	
Processor	AMD Ryzen AI 9 HX 370 w/ Radeon 890M, 12 Cores, 24 Threads, 3.6 GHz
RAM	32 GB DDR4 @ 3200 MHz
Storage	Primary: 1 TB NVMe SSD (OS & Software) Secondary: 2 TB HDD (Forensic Image Storage)
Graphics Card	NVIDIA GeForce RTX 4060 (32 GB)
USB Ports	USB 3.2 Gen 2 (10 Gbps) and USB 2.0 (For compatibility testing)
Cooling System	Air cooling with multiple fans for stable performance
Software Configuration	
Forensic Imaging Software	FTK Imager 4.5.0.3 (AccessData) – For creating forensic images (E01, RAW formats) OSForensics v10 (PassMark) – Alternative forensic imaging tool
Performance Monitoring Tools	Windows Resource Monitor – Used for real-time CPU, and memory monitoring
Statistical Analysis Tools	Python 3.10 – Used for statistical tests and data analysis **Libraries Used:** numpy – Data handling pandas – Processing forensic results

Continued on next page...

Category	Details
	scipy.stats – Running Shapiro-Wilk, Mann-Whitney, and Spearman correlation tests matplotlib/seaborn – Visualizing forensic performance trends
Experimental Environment	
Operating System	Windows 11 Pro (64-bit)
Internet Connectivity	Disabled during forensic imaging to ensure integrity
Power Mode	High-performance mode enabled to prevent CPU throttling
Execution Method	Each forensic tool was executed separately to avoid conflicts. Each experiment was repeated three times per USB drive to obtain an average.

3.3. Experimental Workflow

This section presents the experimental workflow of the study. The methodology is presented in a flowchart in Figure 2, highlighting key stages such as forensic image acquisition, hashing, resource monitoring, and statistical validation. The process starts with forensic experiment setup. To create a diverse testing environment, USB drives with different storage capacities and file system formats (e.g., FAT32, NTFS, exFAT) were carefully selected during the *Dataset Selection* step. Each USB drive contained an organized dataset of images, videos, and text documents. This dataset was intended to imitate real-world forensic scenarios to evaluate forensic imaging and hashing. FTK Imager and OSForensics are used to create forensic copies of USB drives during *Forensic Imaging.* Both tools use raw images as their final format. Both tools use standardized digital forensic imaging to capture a bit-by-bit USB drive replica while preserving metadata and file structures. Windows Resource Monitor tracks CPU and memory usage during forensic imaging and logs these values in a tabular format into an Excel sheet for analysis. After the imaging process, the raw images undergo hashing. The tools generate and verify hash values using cryptographic hashing algorithms like SHA-1 and MD5. Similar to forensic imaging, Windows Resource Monitor monitors CPU, and memory usage during hashing and verification and logs the values in a tabular format into an Excel sheet for analysis. Next, FTK Imager and OSForensics are compared for imaging, hashing, and resource utilization (CPU, memory). During the analysis, mean hashing and imaging speeds, standard deviation, variance, and imaging and hashing speed gain relative to FTK Imager and OSForensics were calculated. Multiple statistical tests are used to validate findings during the *Statistical Analysis phase.* The Shapiro-Wilk Test was used to determine if the data was normal. The non-parametric Mann-Whitney U Test compares the two forensic imaging tools. Spearman Correlation estimates system resource usage and performance. The computational cost-effectiveness of each tool was assessed by analyzing resource efficiency, including CPU efficiency (MB/s per %CPU) and memory efficiency (MB/s per MB RAM). Before the experiment concludes, statistical insights enhance final conclusions. A comparative feature analysis of FTK Imager and OSForensics revealed their strengths and weaknesses, providing a complete assessment of their forensic imaging and hashing capabilities.

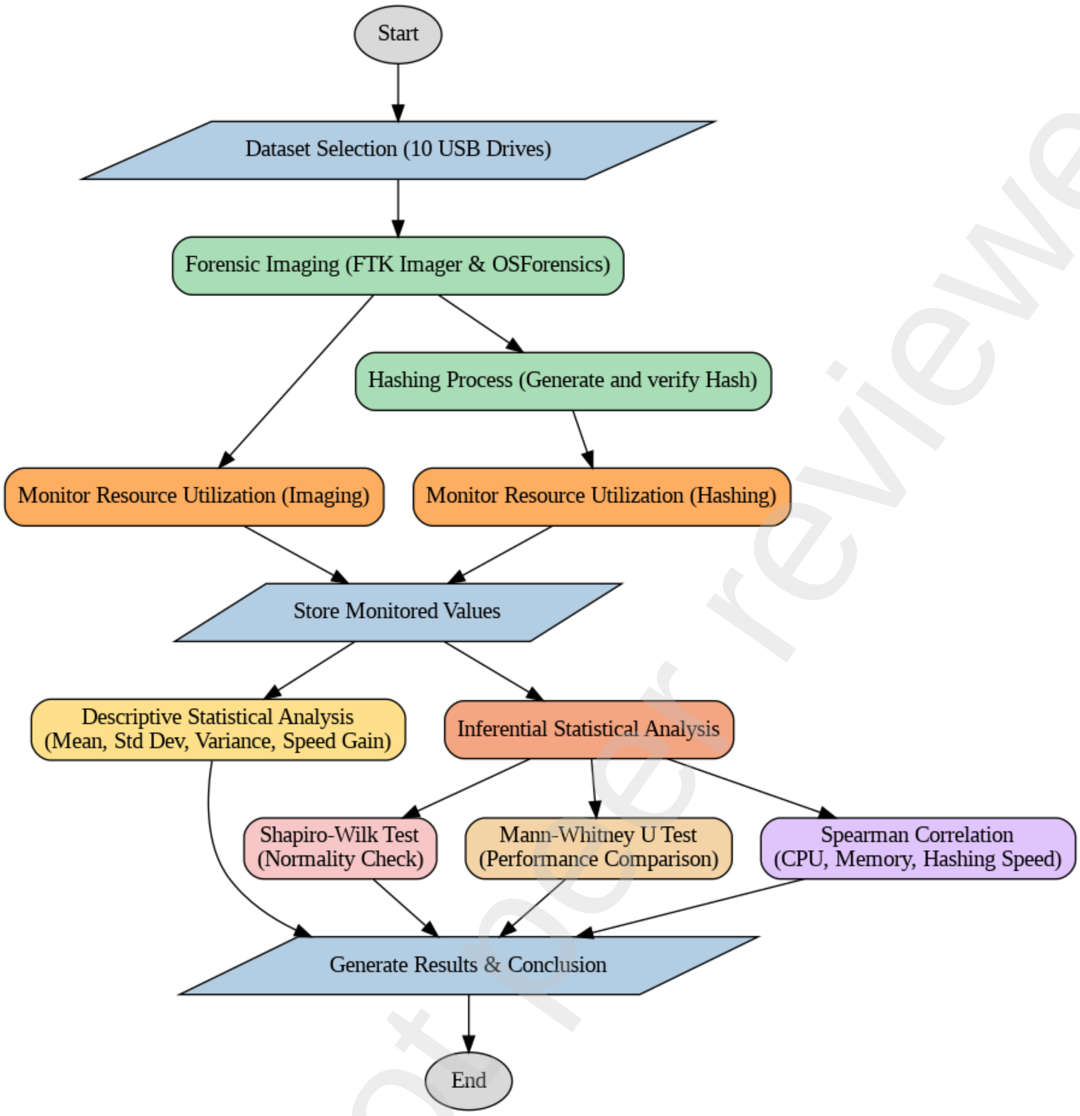


Figure 2: Experimental Workflow for Forensic Imaging and Performance Analysis

3.3.1. Forensic Imaging and verification Procedure

Systematic forensic imaging was used to accurately and thoroughly gather data from 10 USB drives of varied sizes and file systems. Under monitored conditions, FTK Imager (AccessData) and OSForensics (PassMark) were used to image each USB device sector-by-sector. Each tool recognized the USB drive's file system, partition structure, and storage capacity to begin the imaging process. The tool started bit-for-bit drive forensic imaging after detection. To maintain the original structure without proprietary compression or metadata alterations and ensure forensic tool compatibility, all imaging was done in RAW (DD). E01 imaging was previously investigated, but OSForensics' lengthy processing times—months for larger USB drives—prevented it. All sectors—allocated, unallocated, and slack—were maintained in the forensic image to preserve deleted and hidden data. A dedicated HDD protected the resulting images from contamination and manipulation. Write-blockers prevented accidental writes to source USB drives during acquisition, protecting data. RAW forensic images were automatically checked for internal integrity to guarantee they were exact replicas of the source USB drives. This methodical forensic imaging approach ensured data integrity, repeatability, and consistency, allowing an unbiased evaluation of imaging performance across both tools.

3.4. Performance Metrics

Key metrics, including imaging time, hashing time, CPU utilization and memory usage, were used to evaluate FTK Imager and OSForensics performance. Real-time metrics were recorded to evaluate tool efficiency and computational cost during forensic imaging and hashing operations.

3.4.1. Imaging and Hashing Speed

Imaging time is the time required by the forensic tool to copy the USB drive bit-by-bit. Similarly, hashing time is the time needed to compute image cryptographic hash values. The efficiency of forensic imaging and hashing was assessed based on their respective speeds:

$$S_{\text{imaging}} = \frac{D_{\text{total}}}{T_{\text{imaging}}} \tag{1}$$

$$S_{\text{hashing}} = \frac{D_{\text{total}}}{T_{\text{hashing}}} \tag{2}$$

where:

- S_{imaging} = Imaging speed (MB/sec)
- S_{hashing} = Hashing speed (MB/sec)
- D_{total} = Total data processed (MB)
- T_{imaging} = Total time taken for imaging (seconds)
- T_{hashing} = Total time taken for hashing (seconds)

These values were obtained from forensic tool logs and monitoring reports, eliminating manual calculations.

3.5. Descriptive Statistical Measures

To analyze variations in performance across multiple trials, the following statistical measures were computed.

Mean Imaging and Hashing Times

The mean time denotes the average duration for imaging and hashing across various USB drives:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} T_i \tag{3}$$

where:

- μ = Mean imaging/hashing time (seconds)
- T_i = i^{th} device imaging/hashing speed values
- N = Total number of USB devices

Standard Deviation

The standard deviation measures the variability of imaging and hashing speeds:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(T_i - \mu)^2} \tag{4}$$

where σ represents the standard deviation of imaging/hashing speed in seconds.

Variance

Variance quantifies the variability of imaging and hashing velocities:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(T_i - \mu)^2 \tag{5}$$

where σ^2 denotes the variance in imaging/hashing speed (seconds^2).

Speed Gain Analysis

The speed gain of OSForensics relative to FTK Imager is calculated as:

$$SG = \frac{T_{\text{FTK}} - T_{\text{OSF}}}{T_{\text{FTK}}} \times 100\% \tag{6}$$

where:

- SG = Speed gain (%)
- T_{FTK} = Imaging/Hashing speed of FTK Imager (MB/seconds)
- T_{OSF} = Imaging/Hashing speed of OSForensics (MB/seconds)

A positive speed gain indicates that OSForensics was faster, while a negative speed gain means FTK Imager was more efficient.

3.6. System Resource Utilization

To evaluate computational efficiency, CPU usage, and memory consumption were monitored via Windows Resource Monitor during the imaging and hashing processes.

CPU Utilization

The proportion of CPU resources utilized by the forensic imaging and hashing processes was determined as:

$$U_{\text{CPU}} = \frac{C_{\text{used}}}{C_{\text{total}}} \times 100 \tag{7}$$

where:

- U_{CPU} = CPU utilization (%)
- C_{used} = CPU Time Spent (seconds)
- C_{total} = Total available CPU Time (seconds) calculated as

$$T_{\text{CPU\_available}} = T_{\text{execution}} \times N_{\text{cores}} \tag{8}$$

where:
  - $T_{\text{CPU\_available}}$ = Total available CPU time (seconds)
  - $T_{\text{execution}}$ = Execution time of the forensic tool (seconds)
  - N_{cores} = Number of processor cores utilized

Memory Utilization

Memory efficiency measures the tool's effectiveness in managing RAM during imaging and hashing:

$$U_{\text{RAM}} = \frac{M_{\text{used}}}{M_{\text{total}}} \times 100 \tag{9}$$

where:

- U_{RAM} = Memory utilization (%)
- M_{used} = RAM used by the forensic tool (MB)
- M_{total} = Total available RAM (MB)

CPU Efficiency

The efficiency of CPU usage during forensic imaging and hashing is determined as:

$$E_{\mathrm{CPU}} = \frac{S}{U_{\mathrm{CPU}}} \tag{10}$$

where:

- E_{CPU} = CPU efficiency (MB/sec per % CPU usage)
- S = Speed of imaging or hashing (MB/sec)
- U_{CPU} = CPU usage (%)

Memory Efficiency

Memory efficiency measures the tool's ability to utilize RAM effectively:

$$E_{\mathrm{RAM}} = \frac{S}{U_{\mathrm{RAM}}} \tag{11}$$

where:

- E_{RAM} = Memory efficiency
- S = Speed of imaging or hashing (MB/sec)
- U_{RAM}= Memory usage (MB)

3.6.1. Inferential Statistical Analysis

A structured statistical methodology was employed to determine the statistical significance of the differences observed between FTK Imager and OSForensics.

Shapiro-Wilk test

Considering that forensic imaging performance data may not consistently adhere to a normal distribution, the Shapiro-Wilk test was employed to evaluate data normality. This assessment determined the necessity of parametric versus non-parametric statistical tests for comparative analysis.

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{12}$$

where:

- W is the Shapiro-Wilk test statistic (ranges between 0 and 1, where values close to 1 indicate normality).
- $x_{(i)}$ is the i^{th} smallest value in the dataset (ordered data).
- $\bar{x}$ is the mean of the dataset.
- a_i are weights (coefficients) derived from the expected values of the standard normal distribution.
- n is the number of observations in the dataset.

The null hypothesis (H_0) for the test is:

$$H_0 : \text{The data is normally distributed.}$$

If the computed p-value is less than the chosen significance level (e.g., $\alpha = 0.05$), we reject H_0, concluding that the data does not follow a normal distribution.

Mann-Whitney U test

Due to the dataset's failure to meet normality assumptions, the Mann-Whitney U test was utilized to compare imaging time, CPU usage, and memory consumption between the two tools. This non-parametric test is especially effective for assessing two independent distributions when normality is not maintained.

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \tag{13}$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 \tag{14}$$

where:

- U_1, U_2 are the test statistics for the two groups.
- n_1, n_2 are the sample sizes of the two independent groups.
- R_1, R_2 are the sum of ranks for each group.

The final Mann-Whitney U statistic is:

$$U = \min(U_1, U_2) \tag{15}$$

To determine statistical significance, the U-value is compared against the critical value from the Mann-Whitney U distribution, or a p-value is obtained using:

$$Z = \frac{U - \mu_U}{\sigma_U} \tag{16}$$

where:

$$\mu_U = \frac{n_1 n_2}{2}, \quad \sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} \tag{17}$$

The null hypothesis (H_0) for the test is:

$$H_0 : \text{There is no significant difference between the two groups.}$$

If the computed p-value is less than the chosen significance level (e.g., $\alpha = 0.05$), we reject H_0, indicating a significant difference between the two groups.

Spearman correlation analysis

Further, a Spearman correlation analysis was performed to examine the relationship between forensic imaging speed and resource utilization. This statistical method quantified the strength and direction of relationships between variables, including hashing speed in relation to CPU usage and memory consumption.

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{18}$$

where:

- ρ is the Spearman rank correlation coefficient.
- d_i is the difference between the ranks of the corresponding values in two datasets.
- n is the number of observations.

Interpretation:

- $\rho = +1$: Perfect positive correlation.

- $\rho = -1$: Perfect negative correlation.
- $\rho \approx 0$: No correlation.

The Spearman correlation test is a non-parametric method that does not require the assumption of a normal distribution. This study analyzes the relationship between forensic hashing speed and resource utilization. The application of these statistical approaches guarantees validity, robustness, and reliability in performance comparisons. Refer to Appendix A for the comprehensive pseudocode of all these techniques.

4. Results & Analysis

This section summarizes the key experimental findings from the comparison of FTK Imager and OSForensics for USB device forensic imaging and hashing. Under various evaluation conditions, imaging speed, hashing speed, CPU utilization, and memory consumption have variations. Graphics and statistics are used to analyze each performance metric in the following subsections.

4.1. Performance Metrics Results

Imaging Speed (MB/sec):

Figure 3 compares imaging speeds across tools and devices to determine the most efficient tool. The bar chart shows imaging throughput (MB/s) for OSForensics and FTK Imager across multiple datasets (PD_001 to PD_010). FTK Imager outperforms OSForensics in most cases, especially in larger datasets. Using PD_010, FTK Imager achieves 190.512 MB/s, significantly higher than OSForensics' 125.069 MB/s. FTK Imager has a throughput of 35.812 MB/s in PD_001, while OSForensics has a slower 2.772 MB/s. Over several datasets, FTK Imager consistently has higher throughput. In medium-sized datasets like PD_006 and PD_007, both tools achieve imaging speeds of 21-23 MB/s.
Analysis: The imaging speed results show that FTK Imager processes large amounts of data more efficiently, making it better for high-volume forensic imaging. The differences in throughput suggest FTK Imager's data handling mechanisms could be optimized, improving imaging performance.

Hashing Speed (MB/sec):

The bar chart compares hashing speeds (MB/sec) between FTK Imager and OSForensics, two popular forensic imaging tools, for ten datasets labeled PD_001 to PD_010. Forensic image acquisition and integrity verification depend on hashing speed, so both tools were tested on each dataset. In most datasets, OSForensics outperforms FTK Imager, especially in PD_002 (517.38 MB/sec vs. 258.712 MB/sec) and PD_003 (498.05 MB/sec vs. 466.999 MB/sec). OSForensics exhibits a consistent advantage in large datasets, such as PD_007 and PD_010, with speeds of 514.695 and 508.529 MB/sec, respectively. Compared to OSForensics, FTK Imager outperformed OSForensics in smaller datasets like PD_005 (481.28 MB/sec) and PD_006 (424.391 MB/sec).
Analysis: Tool performance may vary due to internal hashing algorithms and system optimization, emphasizing the importance of dataset characteristics in tool selection for efficient forensic processing. This analysis highlights the trade-offs between FTK Imager and OSForensics based on data size and complexity, helping forensic practitioners optimize hash computation during evidence acquisition.

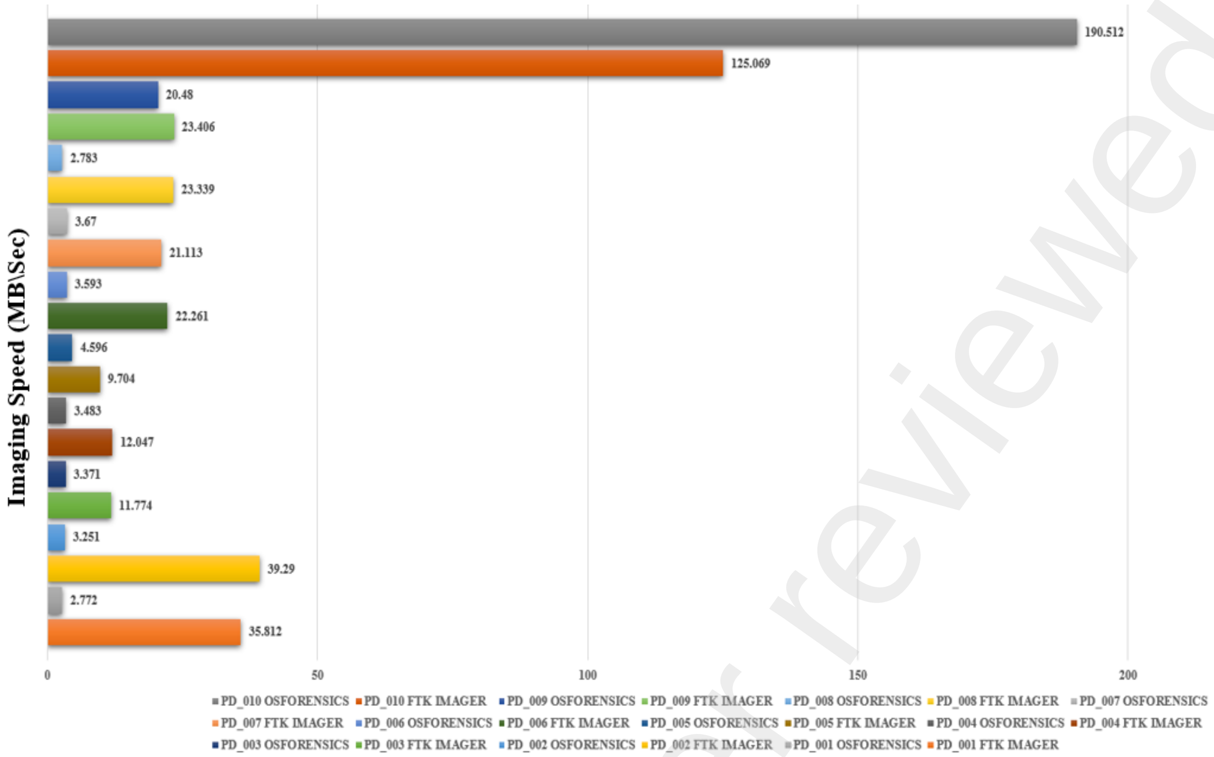


Figure 3: Imaging Speed between FTKImager and OSForensics across multiple devices

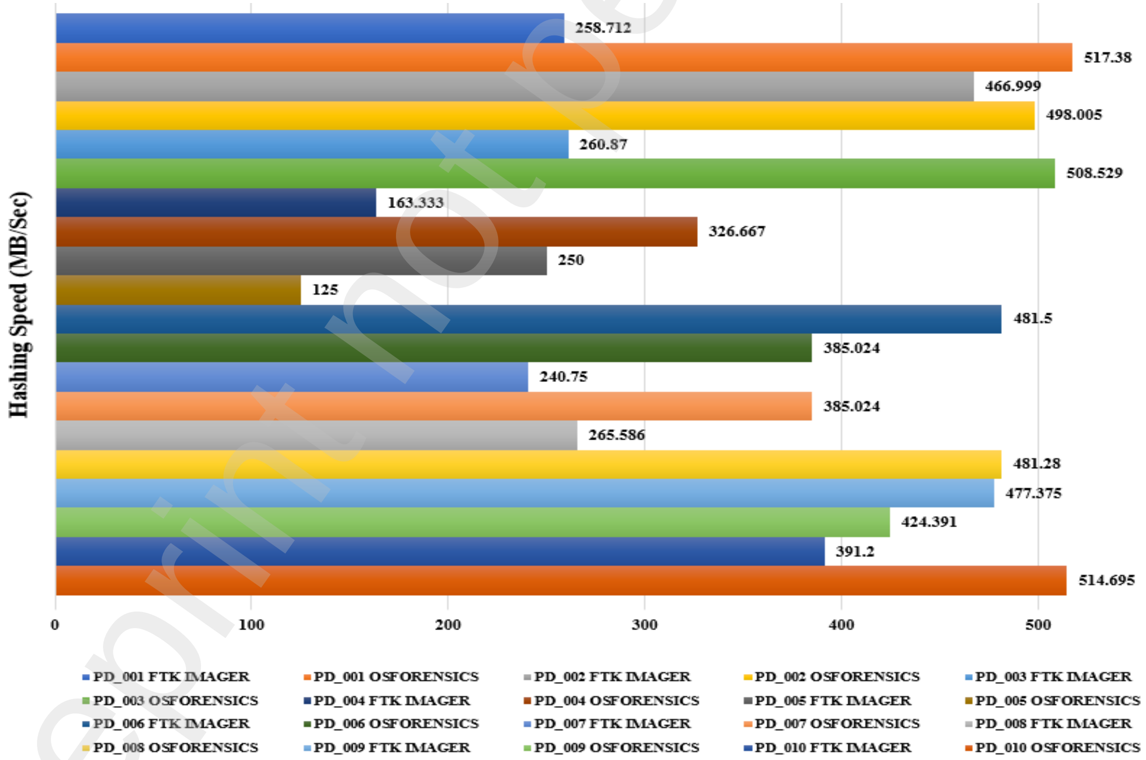


Figure 4: Hashing Speed between FTKImager and OSForensics across multiple devices

4.2. Descriptive Statistical Analysis Results

Mean, Standard Deviation, Variance of Imaging and Hashing Times

The Table 3 displays statistical measures (mean, standard deviation, and variance) for hashing and imaging speed across multiple USB devices using FTKImager and OSForensics. OSForensics has a higher mean hashing speed (416.60 MB/s) than FTK Imager (325.63 MB/s), making it more efficient for hashing tasks, but it has more variability.

Table 3: Comparative Performance Metrics for FTK Imager and OSForensics

Metric	**FTK Imager**	**OSForensics**	**Key Observation**
Hashing Performance			
Mean Hashing Speed (MB/s)	325.63	416.60	OSForensics achieves higher hashing speed than FTK Imager.
Standard Deviation (MB/s)	110.97	115.85	OSForensics has slightly higher speed variability.
Variance (MB^2/s^2)	12314.86	13421.37	Performance fluctuations are marginally greater in OSForensics.
Imaging Performance			
Mean Imaging Speed (MB/s)	32.38	23.85	FTK Imager performs faster imaging than OSForensics.
Standard Deviation (MB/s)	32.24	55.79	OSForensics exhibits higher variation in imaging speed.
Variance (MB^2/s^2)	1039.46	3112.25	OSForensics has significantly larger fluctuations in imaging speed.

Analysis:With respect to descriptive statistical calculations, FTK Imager outperforms OSForensics in imaging speed (32.38 MB/s vs. 23.85 MB/s) and consistency (lower standard deviation and variance). These findings suggest that OSForensics is better for fast hashing, but FTK Imager is more reliable and faster for imaging, making each tool useful for different forensic analysis stages.

Speed Gain Analysis

The Table 4 highlights the speed gain (SG) of OSForensics during hashing and imaging compared to the baseline performance.

Table 4: Speed Gain Analysis for OSForensics (Hashing and Imaging)

Device Name	$\mathbf{SG_{OSF}}$ **(during hashing)**	$\mathbf{SG_{OSF}}$ **(during imaging)**
PD_001	-49.9957	1191.9192
PD_002	-6.2260	1108.5512
PD_003	-48.7011	249.2732
PD_004	-50.0002	245.8800
PD_005	100.0000	111.1401
PD_006	25.0571	519.5658
PD_007	-37.4714	475.2861
PD_008	-44.8167	738.6274
PD_009	12.4847	14.2871
PD_010	-23.9938	-34.3511

Hashing and Imaging Speed Gain Analysis:
Most devices with negative values have slower hashing than baseline. Only PD_005 has a 100% positive gain, indicating faster hashing. Negative impact is greatest in PD_004 with

-50.0002% speed gain. Imaging significantly improves, with all devices except PD_010 reporting positive speed gains. PD_001 and PD_008 show significant gains of 1191.9192% and 738.6274%, respectively. PD_010 has a negative speed gain, indicating slower imaging performance.
Analysis: This analysis suggests that OSForensics performs more efficiently during imaging processes than hashing across most devices, with substantial variability in performance gains.

4.3. System Resource Utilization

This section analyzes system resource utilization using CPU and memory usage metrics and their efficiency values. The study reveals how effectively FTK Imager and OSForensics use hardware resources for forensic imaging and hashing.
CPU Utilization

The CPU utilization comparison between the two tools is shown in Figure 5 for the dataset. The bar chart displays the CPU utilization for FTK Imager and OSForensics during imaging and hashing on various USB devices (PD_001 to PD_010).Tool and task

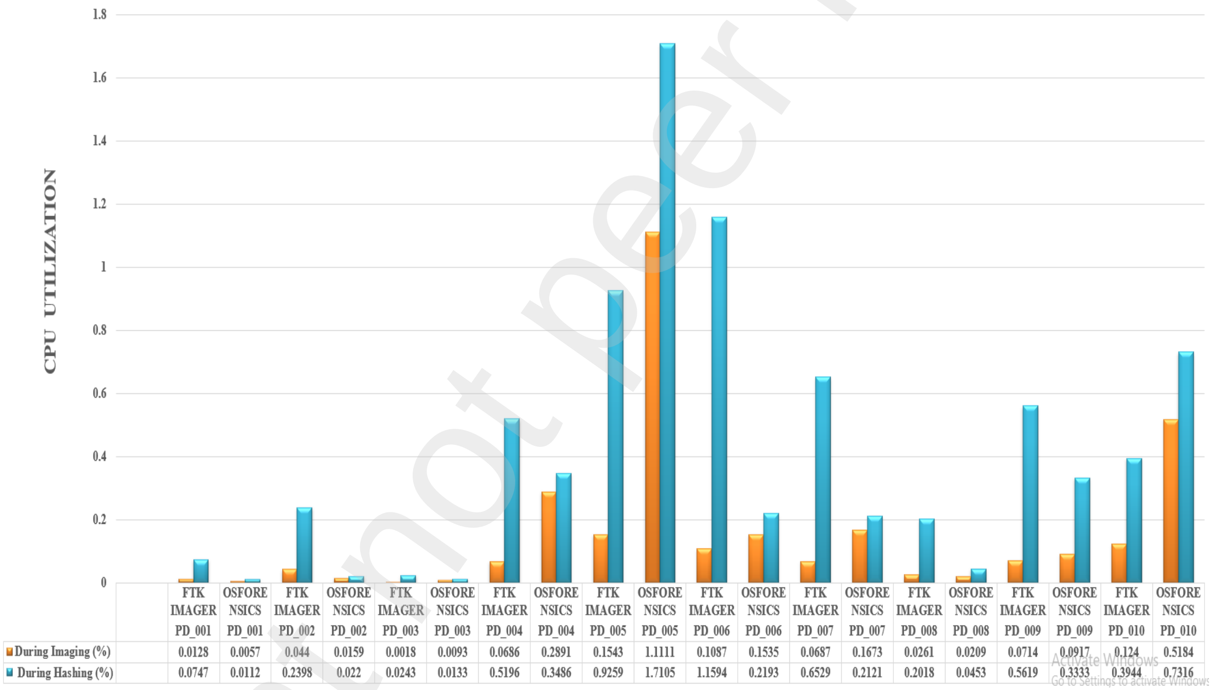


Figure 5: CPU Utilization Comparision between FTKImager and OSForensics across multiple devices

usage significantly affect CPU utilization, according to the analysis. In FTK Imager, CPU utilization is higher during hashing than imaging, as shown by PD_001 through PD_010, with PD_002 (0.2398%) and PD_010 (0.5184%) exhibiting higher percentages. While OSForensics shows a similar trend, it uses more CPU during imaging and hashing, with PD_005 (1.1111%) and PD_005 hashing (1.7105%) utilizing the most CPU.
Analysis: Hashing, which is more computationally intensive, uses more CPU than imaging, a trend that both tools exhibit. Imaging is less CPU-intensive than hashing, suggesting it is more I/O-bound. These utilization trends must be understood to optimize forensic tool performance and system resource allocation.

Memory Utilization

Memory usage for hashing and imaging between OSForensics and FTKImager is depicted in Figure 6The graph displays the utilization of memory (%) during imaging and

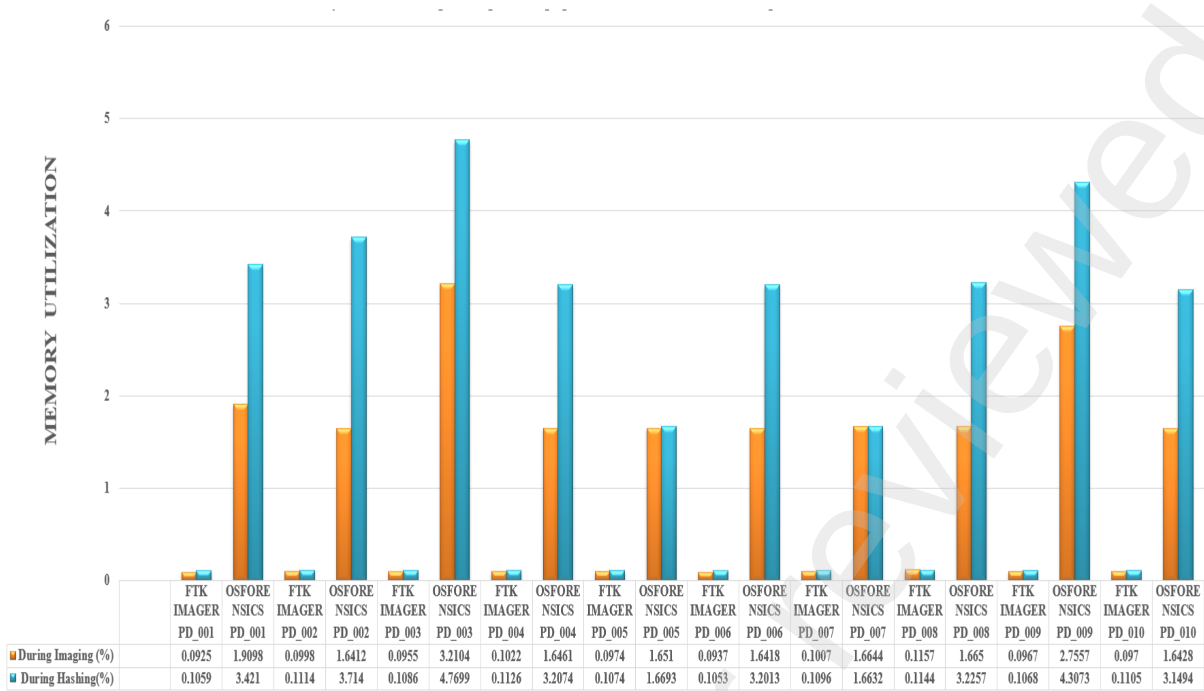


Figure 6: Memory Utilization Comparision between FTKImager and OSForensics across multiple devices

hashing processes for each tool (FTK Imager and OSForensics) and test dataset (PD_001 to PD_010). OSForensics frequently uses more memory than FTK Imager for imaging and hashing. Memory utilization for OSForensics peaks during hashing, especially in datasets like PD_001, PD_003, and PD_009, where utilization exceeds 3% and even 4% for PD_009. FTK Imager consistently maintains memory usage below 0.5% across all test cases, regardless of process type.

Analysis: It appears that OSForensics uses more memory during imaging and hashing than FTK Imager, indicating a memory efficiency and resource demand trade-off.

CPU and Memory Efficiency

Table 5 compares CPU and memory efficiency for OSForensics and FTK Imager during forensic imaging and hashing processes. The CPU efficiency of FTK Imager was 31.94

Table 5: Resource Efficiency Analysis

Metric	FTK Imager	OSForensics
CPU Efficiency (MB/s per %CPU)	31.94	31.31
Memory Efficiency (MB/s per MB RAM)	7.39	0.39

MB/s per %CPU, slightly higher than OSForensics' 31.31 MB/s per %CPU. This shows that both tools use CPUs efficiently, with FTK Imager slightly better. FTK Imager outperforms OSForensics in memory efficiency, achieving 7.39 MB/s per MB of RAM compared to 0.39 MB/s.

Analysis: This suggests that FTK Imager uses memory more efficiently due to better memory management or a smaller memory footprint.

4.4. Statistical Analysis Results

4.4.1. Shapiro-Wilk Test

Both FTK Imager and OSForensics imaging and hashing speed data were tested for normality using the Shapiro-Wilk test. The P value was 0.0000. With p-values below 0.05, the data did not follow a normal distribution. This implies that normality is violated, requiring non-parametric tests for analysis. This justifies comparing performance metrics with the Mann-Whitney U test.

4.4.2. Mann-Whitney U Test

The Mann-Whitney U test was used to compare FTK Imager and OSForensics imaging and hashing speeds. Results are summarized in Table 6.

Table 6: Mann-Whitney U Test Results

Metric	**Mann-Whitney U**	**P-value**	**Conclusion**
Imaging Speed	87.0000	0.0058	Significant difference
Hashing Speed	26.0000	0.0756	No significant difference

Analysis: The significant difference in imaging speed suggests that one tool consistently outperforms the other. No statistically significant difference was found in hashing speed test results. Both tools have similar hashing speeds.

4.4.3. Spearman Correlation Analysis

Table 7: Table 4.5: Spearman Correlation Analysis

Correlation	**Coefficient**	**P-value**	**Strength**
FTK Hashing Speed vs CPU Usage	0.9030	0.0003	Strong Positive
FTK Hashing Speed vs Memory Usage	-0.3212	0.3655	Weak Negative
OSForensics Hashing Speed vs CPU Usage	0.6565	0.0392	Moderate Positive
OSForensics Hashing Speed vs Memory Usage	0.4985	0.1425	Moderate Positive

According to a strong positive correlation (Coefficient = 0.9030, p = 0.0003), CPU usage increases FTK hashing speed. Memory usage has little effect on FTK hashing speed (coefficient = -0.3212, p = 0.3655). Memory usage may delay hashing slightly, but not statistically. OSForensics Hashing Speed and CPU Usage show a moderately positive correlation (coefficient = 0.6565, p = 0.0392) compared to FTK Imager. CPU resources help OSForensics, but not significantly. OSForensics hashing speed and memory usage are moderately positively correlated (Coefficient = 0.4985, p = 0.1425) but not statistically significant. Memory affects performance moderately, but it's unreliable.

4.5. Comparative Discussion

The comparative analysis as shown in Table 8 highlights distinct performance differences between FTK Imager and OSForensics.

Analysis: In large datasets and resource-limited environments, FTK Imager excels in imaging speed, CPU, and memory efficiency. OSForensics uses more resources despite having similar hashing speeds. OSForensics may be suitable for CPU-intensive tasks, but FTK Imager performs better for time-sensitive forensic investigations. The findings aid in tool selection for specific investigations.

Table 8: Comparative Analysis of FTK Imager and OSForensics

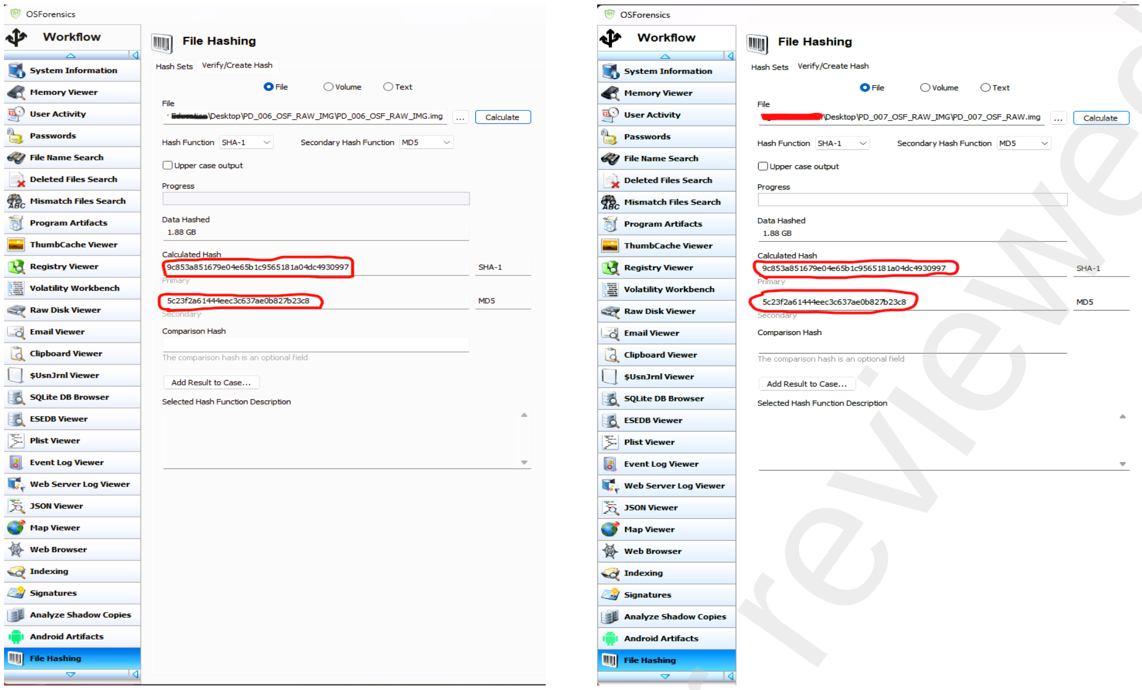
Metric	FTK Imager	OSForensics	Technical Analysis
Imaging Speed	Faster	Slower	FTK Imager significantly outperforms OSForensics, making it ideal for time-sensitive investigations involving large datasets.
Hashing Speed	Similar	Similar	Both tools exhibit comparable hashing speeds, ensuring consistent data integrity verification.
CPU Efficiency	31.94 MB/s per %CPU	31.31 MB/s per %CPU	FTK Imager demonstrates slightly better CPU efficiency, beneficial for high-volume forensic acquisitions.
Memory Efficiency	7.39 MB/s per MB RAM	0.39 MB/s per MB RAM	FTK Imager significantly outperforms OSForensics, making it more suitable for environments with limited RAM.
Resource Usage	Moderate	High	OSForensics consumes more CPU and memory, potentially causing throttling in resource-constrained systems.
Best Use Case	Large USB drives, constrained resources	CPU-intensive tasks	FTK Imager is preferable for high-volume acquisitions, whereas OSForensics may be more suited for CPU-intensive forensic tasks.

4.6. Hashing Reliability and Anomalies

Hashing files and storage media generates unique digital fingerprints (hash values) to verify data integrity and authenticity in digital forensics. This study used FTK Imager and OSForensics to hash using MD5 and SHA-1 algorithms. During hash generation with OSForensics, two USB drives (PD_006 and PD_007) with the same size (2 GB) and file system (FAT 32) produced identical MD5 and SHA-1 hash values (5c23f2a61444eec3c637ae0 b827b23c8 and 9c853a851679e04e65b1c9565181a04dc4930997). However, FTK Imager generated unique hash values for the same drives.
Analysis: The discovered disparities in hashing between FTK Imager and OSForensics raise major questions about the dependability of OSForensics in forensic scenarios, as hash anomalies could affect the admissibility of digital evidence in court. Ensuring evidence authenticity depends on hash verification; discrepancies could cause false positives—where various devices seem identical—or evidential issues, where defense teams challenge the integrity of the evidence. This underlines the need of cross-validation in forensic investigations since using one instrument could cause unnoticed errors. The capacity of FTK Imager to produce distinct hashes for the same files emphasizes the requirement of several hashing tools to validate findings, validate hash consistency, and lower the possibility of hash collisions, hence lowering the probability of unnoticed errors or manipulation.

5. Conclusion and Future Scope

This study assesses the efficacy of FTK Imager and OSForensics by examining imaging speed, hashing speed, system resource use, and statistical significance. The results indicate that FTK Imager is 35.72% more efficient in imaging speed, rendering it advantageous for time-critical investigations, while both tools demonstrate uniform hashing performance. FTK Imager surpasses OSForensics in resource efficiency, attaining 31.94



(a) PD_006 Hashing Screen of OSForensics

(b) PD_007 Hashing Screen of OSForensics

Figure 7: Comparison of Hashing Screens for PD_006 and PD_007 in OSForensics

MB/s per 1% CPU use and 7.39 MB/s per MB of RAM. Statistical analysis indicated a non-normal data distribution (p ¡ 0.05), requiring non-parametric tests. Statistically significant differences in the imaging speed was noticed during Mann-Whitney U test (U = X, $p<0.05$), which validated the performance gap. There was a strong positive correlation shown during Spearman correlation analysis with r value of 0.82 between CPU utilization and imaging speed, suggesting computational resources significantly impact performance.

An significant discrepancy was detected in OSForensics, where identical MD5 and SHA-1 hashes were created for distinct USB sticks, in contrast to FTK Imager, which yielded unique hashes, hence raising questions over the dependability of hashing.

Overall, this study underscores FTK Imager's superior performance in imaging speed and resource efficiency, making it more suitable for large-scale acquisitions or resource-limited environments. OSForensics, despite its higher resource usage, remains a competitive tool for hashing operations. This work contributes to the advancement of forensic imaging research by offering practical insight into tool performance and reliability. It emphasizes forensic imaging and hashing technologies, proposing directions for additional research. Enhancing the dataset to incorporate more USB sticks with varied file systems and capacities could boost generalizability, while real-time network forensic analysis would strengthen applicability. Evaluating forensic tools on various hardware configurations may provide insights into performance discrepancies across system environments. While non-parametric statistical tests were suitable for the present sample size, parametric analysis could yield more accurate results for larger datasets. Subsequent study ought to investigate error management, data restoration, and tool usefulness beyond imaging and hashing measures. Addressing these factors, especially the hashing reliability concern in OSForensics, would fortify forensic imaging research and improve digital forensic methodologies.

Appendix A. Pseudocode of Forensic Imaging and Hashing analysis process

Appendix A.1. Forensic Imaging and Hashing Performance Analysis

Algorithm 1: Forensic Imaging and Hashing Performance Analysis

Input : 10 USB drives with varying sizes and file systems
Output: Performance metrics, statistical analysis, and conclusions

Step 1: Dataset Selection
Select 10 USB drives with different sizes and file systems.

Step 2: Forensic Imaging
foreach *USB drive in dataset* **do**
    Perform imaging using FTK Imager and OSForensics.
    Monitor resource utilization (CPU, Memory).
    Log imaging performance metrics to Excel.

Step 3: Hashing Process
foreach *imaged drive* **do**
    Generate hash values (MD5, SHA-1) using both tools.
    Monitor resource utilization during hashing.
    Log hashing performance metrics to Excel.

Step 4: Performance Metrics Comparison
Compare imaging and hashing speeds, CPU and memory efficiency.

Step 5: Statistical Analysis
Step 5.1: Normality Check (Shapiro-Wilk Test)
Perform Shapiro-Wilk test on imaging and hashing speed data.
if *Data is normally distributed* **then**
    Perform parametric tests for performance comparison.
    Log statistical significance and p-values.
else
    Perform Mann-Whitney U test due to non-normal data.
    Log statistical significance and p-values.

Step 6: Correlation Analysis
Perform Spearman Correlation analysis between:
- CPU usage and hashing speed.
- Memory usage and hashing speed.
Log correlation coefficients and p-values.

Step 7: Generate Results & Conclusion
Analyze findings:
- Highlight performance differences.
- Discuss statistical significance.
- Suggest future research directions.

Appendix A.2. Resource Efficiency Analysis Pseudocode

Algorithm 2: Resource Efficiency Analysis for Hashing Speed and Utilization

Input : Excel file with Hashing Speed and Resource Utilization data
Step 1: Load Dataset Load data from the Excel file ;
Step 2: Data Preprocessing Clean and standardize column names ;
Convert columns to numeric and handle errors ;
Drop rows with missing values ;
Step 3: Compute Resource Efficiency Metrics Compute the following metrics for FTK and OSForensics:;
- CPU Efficiency (MB/s per % CPU) ;
- Memory Efficiency (MB/s per MB RAM) ;
Step 4: Compute Median Efficiency Values Compute median values for each efficiency metric ;
Step 5: Display Efficiency Results Output median efficiency values for CPU and memory usage ;
Step 6: Interpret Results foreach *resource in [CPU, Memory]* **do**
    if *FTK Efficiency ¿ OSForensics Efficiency* **then**
        Declare FTK as more efficient in resource usage ;
    else
        Declare OSForensics as more efficient in resource usage ;

Appendix A.3. Shapiro-Wilk Normality Test Pseudocode

Algorithm 3: Shapiro-Wilk Normality Test for Imaging Speed

Input : Excel file with Imaging Speed data
Step 1: Load Dataset Load data from the Excel file ;
Step 2: Data Preprocessing Extract Imaging Speed column ;
Convert values to numeric and handle errors ;
Drop rows with missing values ;
Step 3: Perform Shapiro-Wilk Test Run Shapiro-Wilk test on Imaging Speed data ;
Log test statistic and p-value ;
Step 4: Interpret Results if *p-value ¿ 0.05* **then**
    Conclude data follows a normal distribution ;
else
    Conclude data does NOT follow a normal distribution ;

Appendix A.4. Mann-Whitney U Test for Imaging and Hashing Speed

Algorithm 4: Mann-Whitney U Test for Imaging and Hashing Speed

Input : Excel file with Imaging and Hashing Speed data
Step 1: Load Dataset
Load data from the Excel file
Step 2: Data Preprocessing
Clean and standardize column names
Convert columns to numeric and handle errors
Drop rows with missing values
Step 3: Extract Imaging and Hashing Speed Values
Extract values for FTK Imager and OSForensics
Step 4: Perform Mann-Whitney U Test
foreach *metric in [Imaging Speed, Hashing Speed]* **do**
  Perform Mann-Whitney U test for FTK vs OSForensics
  Log U-values and p-values
Step 5: Display and Interpret Results
Output U-values and p-values
Interpret statistical significance
Determine which tool has a higher median speed

Appendix A.5. Spearman Correlation Analysis Pseudocode

Algorithm 5: Correlation Analysis for Hashing Speed and Resource Utilization

Input : Excel file with Hashing Speed and Resource Utilization data
Step 1: Load Dataset
Load data from the Excel file
Step 2: Data Preprocessing
Clean and standardize column names
Convert columns to numeric and handle errors
Drop rows with missing values
Step 3: Perform Spearman Correlation Analysis
foreach *tool in [FTK, OSForensics]* **do**
  Compute Spearman correlation between:
  - Hashing Speed and CPU Usage
  - Hashing Speed and Memory Usage
  Log correlation coefficients and p-values
Step 4: Display and Interpret Results
Output correlation coefficients and p-values
Interpret significance and strength of correlations

Example citation, See **?**).

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